

Revision — surface areas and volumes of all the solids

Nine formulas, one page, and the three questions that decide which of them a problem wants.

LESSON 65–66

2 × 40 MIN

GEOMETRY 11, FULL MENSURATION REVIEW

IGX 13.X FULL REVIEW

LESSON CLOCK

15'

20'

25'

25'

15'

The nine formulas	15 min
The three questions	20 min
Compound and hollow	25 min
Scale factors	25 min
A mixed set	15 min

LEARNING OBJECTIVES

- Recall and apply every surface-area and volume formula of the two-year course.
- Decide from the wording which measure a problem wants.
- Handle compound and hollow solids, and scale factors.
- Work a mixed set at examination pace.

TEXTBOOK

National	pp. 335–344
Cambridge	Core & Extended, pp. 397–404
Workbook	Revision set G4

01

Explanation

The nine formulas

SOLID	VOLUME	LATERAL / CURVED	TOTAL
prism	Bh	Ph	$2B + Ph$
cylinder	$\pi r^2 h$	$2\pi r h$	$2\pi r(r + h)$
pyramid (regular)	$\frac{1}{3} Bh$	$\frac{1}{2} Pm$	$B + \frac{1}{2} Pm$
cone	$\frac{1}{3} \pi r^2 h$	$\pi r l$	$\pi r(r + l)$
frustum of a cone	$\frac{\pi h}{3} (R^2 + Rr + r^2)$	$\pi l(R + r)$	+ two circles
sphere	$\frac{4}{3} \pi R^3$	$4\pi R^2$	$4\pi R^2$
hemisphere	$\frac{2}{3} \pi R^3$	$2\pi R^2$	$3\pi R^2$

THREE PATTERNS WORTH MORE THAN THE NINE LINES

- Every **pointed** solid is a third of the matching flat-topped one: $\frac{1}{3}$ for pyramid and cone.
- Every **lateral** area is “perimeter times a length, halved if pointed”: Ph and $\frac{1}{2} Pm$; $2\pi r h$ and $\pi r l$.
- Every **curved** area of a cone or pyramid uses the **slant**; every volume uses the **height**.

Learn the three patterns and the nine formulas rebuild themselves.

The three questions

BEFORE CHOOSING A FORMULA

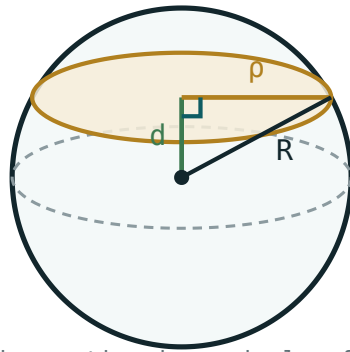
1. **Volume or area?** The words are “hold”, “fill”, “weigh” against “paint”, “cover”, “label”.
2. **Which faces?** Open or closed, solid or hollow, standing on the ground or not.
3. **Which length?** Height for volume, slant for curved area, radius not diameter.

WORDING	FORMULA
“a label around a tin”	$2\pi rh$
“a lid for a tin”	πr^2
“the canvas of a conical tent, no floor”	πrl
“the concrete in a solid pyramid”	$\frac{1}{3} Bh$
“the water in a hemispherical bowl”	$\frac{2}{3} \pi R^3$
“the glass in a hollow sphere”	$\frac{4}{3} \pi (R^3 - r^3)$

Compound and hollow

$$V_{\text{compound}} = V_1 + V_2 \quad S_{\text{compound}} = S_1 + S_2 - 2 \times (\text{shared face})$$

OBJECT	VOLUME	OUTSIDE AREA
silo: cylinder + hemisphere	$\pi r^2 h + \frac{2}{3} \pi r^3$	$2\pi rh + 2\pi r^2 (+ \text{base?})$
ice-cream: cone + hemisphere	$\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$	$\pi rl + 2\pi r^2$
pipe: cylinder – cylinder	$\pi(R^2 - r^2)L$	$2\pi(R + r)L + 2 \text{ rings}$
hollow sphere	$\frac{4}{3} \pi (R^3 - r^3)$	$4\pi(R^2 + r^2)$



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

A plane cut through a solid of revolution — the shared face is always a circle.

THE TWO SHARED CIRCLES VANISH, NOT ONE

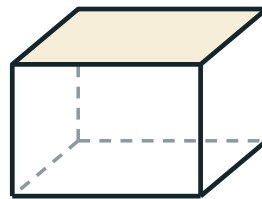
When a hemisphere sits on a cylinder, the cylinder's top circle and the hemisphere's flat face are both internal. Subtracting only one of them leaves πr^2 too much every time.

Scale factors, and a mixed set

$length \times k \quad area \times k^2 \quad volume \times k^3$



$k = 1$



$k = 2$

lengths $\times k$ · areas $\times k^2$ · volumes $\times k^3$

One scale factor, three different powers.

Read backwards this is just as useful: from a volume ratio take the cube root to get k , then square it for areas.

#	QUESTION	ANSWER
1	Volume of a cone, $r = 6$, $h = 8$	96π
2	Total area of that cone	96π
3	Volume of a sphere of radius 9	972π
4	Surface area of that sphere	324π
5	Volume of a square pyramid, base 10, height 12	400
6	Its lateral area	260
7	Frustum, $R = 5$, $r = 3$, $h = 6$	98π
8	Two similar solids, areas 9 : 25: volume ratio	27 : 125

QUESTION 2 IS THE TRAP

A cone with $r = 6$ and $h = 8$ has $l = 10$, so its total area is $\pi r(r + l) = 96\pi$ — numerically equal to its volume, purely by coincidence of these numbers. Recognising that the two answers **should** differ in units is the check that catches a formula used in the wrong place.

02 Worked examples

EXAMPLE 1 A cone has $r = 6$ and $h = 8$. Find its volume and total surface area.

$$(1) \quad l = \sqrt{36 + 64} = 10$$

$$(2) \quad V = \frac{1}{3}\pi(36)(8) = 96\pi$$

Height.

$$(3) \quad S = \pi(6)(6 + 10) = 96\pi$$

Slant.

$$(4) \quad \text{Equal numbers, different units — cm}^3 \text{ against cm}^2.$$

ANSWER

$96\pi \text{ cm}^3$ and $96\pi \text{ cm}^2$

EXAMPLE 2 A hollow glass sphere has outer radius 10 cm and inner 9 cm. Find the volume of glass.

$$(1) \quad V = \frac{4}{3}\pi(1000 - 729)$$

$$(2) \quad = \frac{4}{3}\pi(271)$$

$$(3) \quad \approx 1135 \text{ cm}^3.$$

ANSWER

$$\frac{1084\pi}{3} \approx 1135 \text{ cm}^3$$

EXAMPLE 3 Two similar cones have surface areas in the ratio 9 : 25. Find the ratio of their volumes.

$$① \quad k^2 = \frac{25}{9} \Rightarrow k = \frac{5}{3}$$

Square root first.

$$② \quad k^3 = \frac{125}{27}$$

$$③ \quad 27 : 125$$

ANSWER

27 : 125

EXAMPLE 4 An ice-cream is a cone of radius 3 and height 4 with a hemisphere on top. Find its volume and outside surface area.

$$① \quad V = \frac{1}{3} \pi (9)(4) + \frac{2}{3} \pi (27) = 12\pi + 18\pi = 30\pi$$

$$② \quad l = 5, \text{ so cone curved } \pi(3)(5) = 15\pi.$$

No base circle — it is joined.

$$③ \quad \text{Hemisphere curved } 2\pi(9) = 18\pi.$$

Its flat face is joined too.

$$④ \quad S = 33\pi \approx 104 \text{ cm}^2.$$

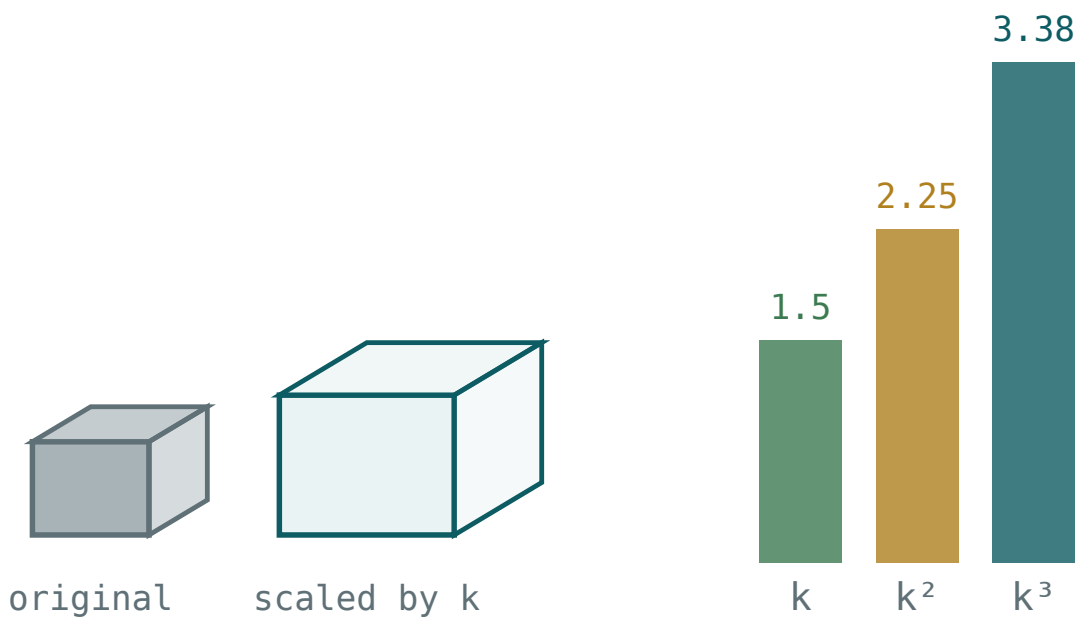
ANSWER

$30\pi \approx 94.2 \text{ cm}^3$; $33\pi \approx 104 \text{ cm}^2$

03 Interactive model

Put the nine formulas on the board, then rub out three at random and have the class rebuild them from the three patterns.

One factor, three powers



k 1.5

length $\times$ 1.5area $\times k^2$ 2.25volume $\times k^3$ 3.38

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Cylinder	Silindr	Цилиндр
Pyramid	Piramida	Пирамида
Cone	Konus	Конус
Frustum	Kesik jism	Усечённое тело
Sphere	Sfera	Сфера
Lateral surface	Yon sirt	Боковая поверхность
Total surface	To'la sirt	Полная поверхность
Scale factor	O'xshashlik koeffitsiyenti	Коэффициент подобия
Compound solid	Murakkab jism	Составное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Volume of a pyramid:

Bh

$\frac{1}{3} Bh$

$\frac{1}{2} Bh$

$3Bh$

Curved area of a cone:

πr^2

$\pi r l$

$\pi r h$

$2\pi r l$

Volume of a cone:

$\pi r^2 h$

$\frac{1}{3} \pi r^2 h$

$\frac{1}{3} \pi r^2 l$

$\pi r l$

Total area of a hemisphere (solid):

$$2\pi R^2$$

$$3\pi R^2$$

$$4\pi R^2$$

$$\pi R^2$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\pi h(R^2 - r^2)$$

$$\frac{\pi h}{2}(R + r)$$

Areas in the ratio 9 : 25 give volumes in:

$$9 : 25$$

$$27 : 125$$

$$3 : 5$$

$$81 : 625$$

Two joined faces of a compound solid:

both count

one counts

neither counts

they are halved

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a prism, $B = 20$, $h = 7$

ANSWER 140

2. Volume of a cylinder, $r = 3$, $h = 5$

ANSWER 45π

3. Volume of a cone, $r = 6$, $h = 8$

ANSWER 96π

4. Slant of that cone

ANSWER 10

5. Volume of a sphere, $R = 9$

ANSWER 972π

6. Surface area of that sphere

ANSWER 324π

7. Volume of a pyramid, $B = 100$, $h = 12$

ANSWER 400

Medium · the standard question

7 PROBLEMS

1. Total area of a cone, $r = 6$, $h = 8$

ANSWER 96π

2. Lateral area of a square pyramid, base 10, height 12

ANSWER 260

3. Frustum volume, $R = 5$, $r = 3$, $h = 6$

ANSWER 98π

4. Total area of a cylinder, $r = 3$, $h = 5$

ANSWER 48π

5. Total area of a solid hemisphere, $R = 4$

ANSWER 48π

6. Areas 9 : 25: the volume ratio

ANSWER 27 : 125

7. Volumes 8 : 27: the area ratio

ANSWER 4 : 9

Hard · stretch and reason

7 PROBLEMS

1. Hollow glass sphere, $R = 10$, $r = 9$: volume of glass

ANSWER $\approx 1135 \text{ cm}^3$

2. Ice-cream, cone $r = 3$, $h = 4$ plus hemisphere: volume and outside area

ANSWER $30\pi; 33\pi$

3. Silo, cylinder $r = 4$, $h = 12$ plus hemisphere: volume

ANSWER $\frac{704\pi}{3} \approx 737$

4. Pipe $R = 6$, $r = 4$, $L = 150$: volume of metal

ANSWER $3000\pi \approx 9425$

5. A cone of radius 9 cut halfway up: volume of the frustum as a fraction

ANSWER $\frac{7}{8}$

6. A sphere inscribed in a cylinder: the ratio of total areas

ANSWER $2 : 3$

7. A cube and a sphere of equal volume: the ratio of surface areas (cube : sphere)

ANSWER $6 : \sqrt[3]{36\pi} \approx 1 : 0.806$

07 Homework

Homework — 5 tasks

Answer the three questions in writing before each calculation; that is the whole revision.

- Write the nine formulas from memory, then check them and mark any you got wrong.
- A cone has radius 5 cm and height 12 cm. Find its volume, curved area and total area.
- A hollow metal sphere has outer radius 8 cm and inner 7 cm. Find the volume of metal and its mass at 7.8 g/cm^3 .

4. Two similar solids have volumes 64 and 125 cm^3 . Find the ratio of their heights and of their surface areas.
5. A grain hopper is a cylinder of radius 2 m and height 5 m with a cone of height 3 m below it. Find its capacity in litres.